



Study on the evolution processes of keyhole and melt pool in different laser welding methods for dissimilar materials based on a novel numerical model

Yuewei Ai^{a,b,*}, Chenglong Ye^{a,b}, Jiabao Liu^{a,b}, Ming Zhou^{a,b}

^a School of Traffic and Transportation Engineering, Central South University, Changsha 410075, PR China

^b Key Laboratory of Traffic Safety on Track of Ministry of Education, Central South University, Changsha 410075, PR China

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ABSTRACT

A novel numerical model based on a combined heat source and an improved recoil pressure formula is established to explore the effect of laser beam oscillation on the evolution processes of keyhole and melt pool in laser welding of Q235 low carbon steel and 316L stainless steel dissimilar materials stake welded T-joints. The welding processes in conventional laser welding (CLW) and oscillating laser welding (OLW) are solved by the numerical model. The evolution processes of morphologies of keyhole and melt pool and flow behaviors of molten metal in CLW and OLW are compared. It is found that the morphologies of keyhole and melt pool in OLW are significantly different from that in CLW. The keyhole depths and volumes and melt pool depths and volumes in CLW and OLW are all increased firstly and then kept in the steady state. Comparing with CLW, the keyhole depth and volume and melt pool depth and volume in OLW all indicate the periodic fluctuation characteristics with large range and roughly the same period as the laser beam oscillation. At the steady stage of OLW, the flow velocity magnitude distribution uniformity of molten metal is increased and flow direction of molten metal is more complicated with many large area vortices formation, which are beneficial for suppressing the pore defect in weld and improving the welding quality.

1. Introduction

The stake welded T-joints which are joined by face plate and web plate perpendicularly are as the typical structures with the lightweight and high-strength advantages, which has been extensively used in ships, bridges, rocket and other fields [1–4]. To meet the diversified requirements, the stake welded T-joints with different performances can be customized by choosing dissimilar materials as face plate and web plate. As a green and advanced joining technology, laser welding is an ideal joining method to achieve the joining of face plate and web plate in the stake welded T-joints due to high energy density, high penetration capability and low distortion [5,6]. However, solute segregation, pores, spatters and other welding defects are easy to be generated in the welding of dissimilar materials stake welded T-joints owing to welding process instability and dissimilar materials property differences, which will affect the welding quality of stake welded T-joints greatly.

Dissimilar materials laser welding as a challenge has been focused in the welding field recently. Oliveira et al. [7] conducted CoCrFeMnNi high entropy alloy and 316L stainless steel laser welding. They found that the as-cast substrate hardness in the fusion zone was increased,

which was probably caused by the solid solution strengthening due to dissimilar materials mixing and carbon incorporation from 316L stainless steel. Rossini et al. [8] carried out the laser welding experiments with four kinds of high strength steels. It was found that the segregation of Mn element might weaken the mechanical properties of dissimilar materials butt joints and lead to the failure in the fusion zone due to the intergranular fracture along the prior austenite grain boundaries. Wu et al. [9] obtained the SA553-SUS304 dissimilar materials joints by laser welding. The extremely inadequate mixture of dissimilar materials in the partial melting zone near SA553 interface was found under the condition of high cooling rate, which resulted in the formation of martensite with different morphologies and compositions. Yan et al. [10] investigated Fe/Al laser welding and observed that Fe₂Al₅, FeAl₃, FeAl and other Fe-Al intermetallic compounds were formed at Fe/Al interface due to composition segregation, which led to the decrease in plastic and ductility of welds. Chen et al. [11] obtained the 316L/GH909 dissimilar materials welded joints under the process conditions with the same linear energy. It was found that smaller temperature gradient at edge of melt pool was formed under low laser power and low welding speed, which resulted in more uniform distributions of elements and

* Corresponding author at: School of Traffic and Transportation Engineering, Central South University, Changsha 410075, PR China.

E-mail address: aiyuewei1@126.com (Y. Ai).

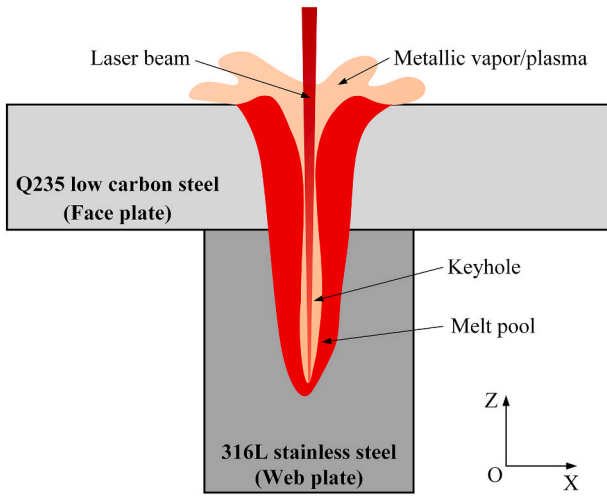


Fig. 1. Schematic diagram of welding experiments.

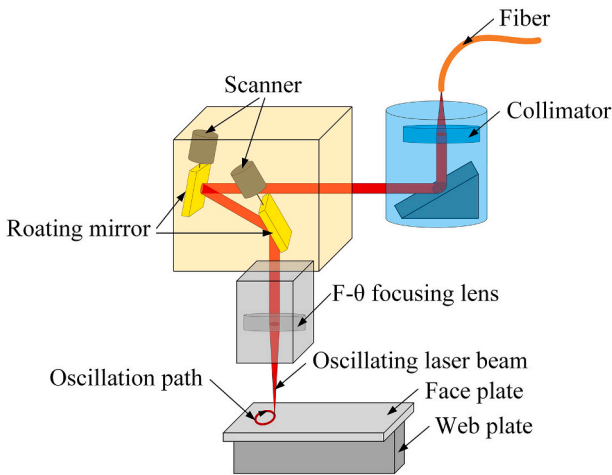


Fig. 2. Schematic diagram of galvanometer scanner.

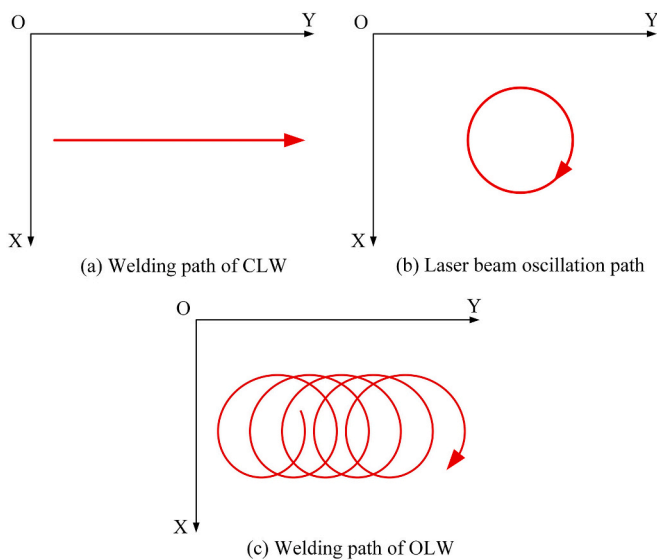


Fig. 3. The path schematic diagram.

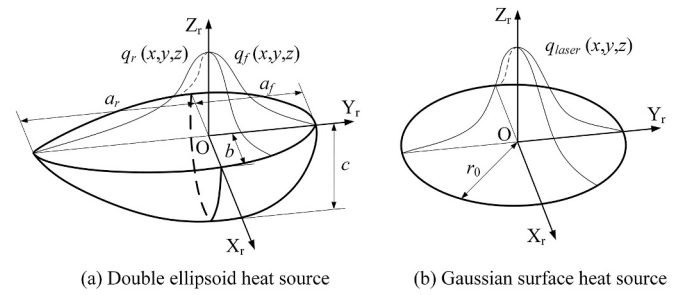


Fig. 4. Models of double ellipsoid heat source and Gaussian surface heat source.

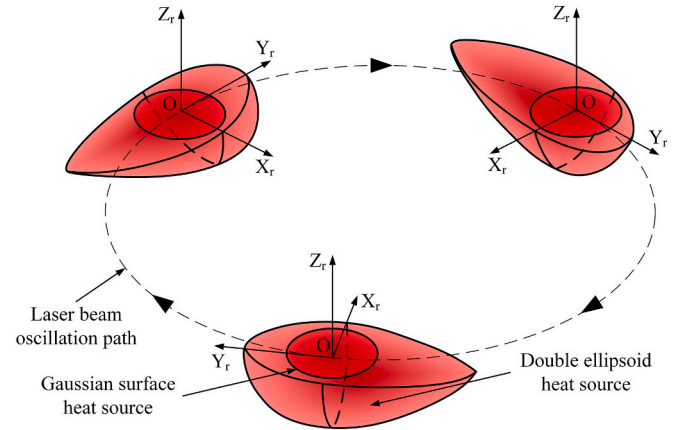
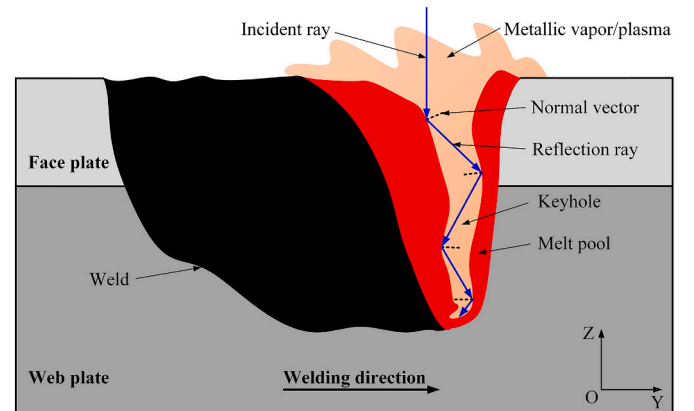
Fig. 5. Combined heat source model which realizes rotation about the heat source central axis (Z_r) and the revolution along the laser beam oscillation path in OLW.

Fig. 6. Ray multiple reflections process.

microstructure and better welded joint tensile property. Esfahani et al. [12] studied melt pool dynamic behaviors in low carbon steel and stainless steel laser welding by numerical calculation. They found that joint properties were influenced by alloy element concentration and weld homogeneity. The melt pool dynamic behaviors and alloy mixture in fusion zone were affected by temperature gradient and surface tension. Chatterjee et al. [13] analyzed microstructure in the Ti/Ni laser welding and found that weld macroscopic morphology asymmetry was caused by dissimilar materials thermophysical property differences. The macro-segregation was formed since molten metal flow behaviors induced by the density difference of material components. Chen et al. [14] analyzed the effect of heat source location on welded joint

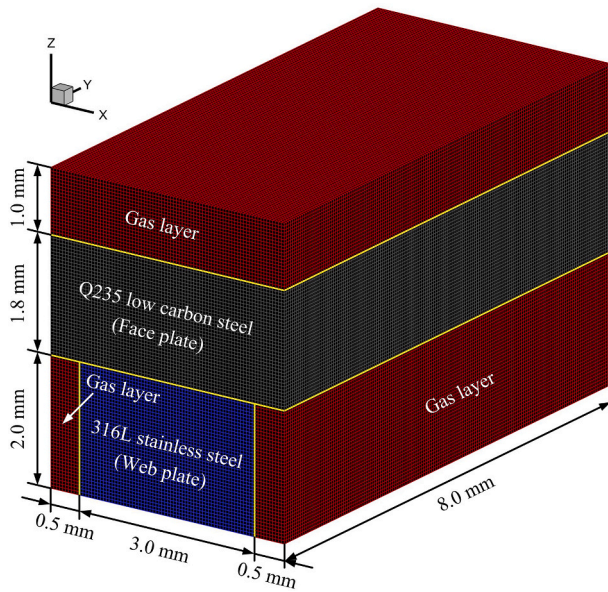


Fig. 7. The mesh model.

mechanical properties in dissimilar materials hybrid laser-TIG welding. Their results showed that the heat source location had great influence on weld morphology, microstructure and macro-segregation and welded joint mechanical properties. Halim et al. [15] discussed the transport phenomena in Al/Mg laser welding and found that molten metal dynamic behaviors which were influenced by temperature gradient variation and inter-diffusion between dissimilar materials had great effects on the molten metal convection and melt pool morphology.

As a new welding method with the potential to improve welding quality, oscillating laser welding (OLW) has received extensive attention. Ai et al. [16] discussed the effects of oscillation parameters on energy distribution and dynamic behaviors in circular shaped OLW. They found that increasing the oscillation amplitude or frequency could improve energy distribution uniformity. The stability of keyhole, the distributions of temperature and flow velocity as well as the morphology of melt pool could be improved by adjusting the oscillation parameters. Li et al. [17] explored the effect of oscillation path on melt pool dynamic behaviors. The melt pool with the lowest temperature gradient and the most complicated vortices and the stablest keyhole were obtained in

circular shaped OLW, which helped reduce the pore defect in the weld. Liu et al. [18] revealed the mechanism of pore reduction in sinusoidal shaped OLW of 7075 aluminum alloy. They found that the keyhole was more stable and the pores in the weld were fewer in OLW than that in conventional laser welding (CLW), which was caused by increasing keyhole diameter and weakening surface tension. Xia et al. [19] obtained the weld with fewer pores by OLW. It was because oscillation process could improve energy distribution, reduce peak temperature of melt pool and weaken the effect of molten metal flow behaviors on keyhole. Zhou et al. [20] studied Ti/Al OLW and found that pore and crack defects were reduced, weld width at interface was increased and Al to Ti diffusion was weakened, which could improve the weld quality. Liu et al. [21] discussed the effect of laser beam oscillation on pore defect suppression in medium-thick high-magnesium aluminum alloy laser welding. They found that pores were reduced due to keyhole opening expansion and suppression of keyhole collapse in infinity shaped OLW. Han et al. [22] proposed a novel welding strategy to produce ultra-fine equiaxed grains in Al-Li OLW. The OLW could reduce pore defect and facilitate Zr element diffusion to improve the uniformity of microstructure distribution. Chen et al. [23] carried out D406A ultra-high strength steel OLW. Their observed results indicated that the small bubbles were formed after the large bubbles hit by laser beam and then small bubbles were probably captured by keyhole and escaped from melt pool with metallic vapor.

From the previous researches, it can be found that the OLW under the suitable process conditions has the potential to improve the uniformities of element, microstructure and temperature distributions and reduce pore defect. Therefore, the circular shaped OLW is employed for the Q235 low carbon steel and 316L stainless steel laser welding. A novel numerical model based on a combined heat source and an improved recoil pressure formula which can achieve the high calculation accuracy is developed to explore the evolution processes of keyhole and melt pool in CLW and OLW, which is of great importance for improving dissimilar materials stake welded T-joint welding quality.

2. Experiments

In the laser welding experiments, Q235 low carbon steel and 316L stainless steel are selected as face plate and web plate, respectively, as shown in Fig. 1. The contact surfaces of face plate and web plate in the stake welded T-joint assembly are milled by a milling machine to minimize the gap of T-joint. The dimensions of face plate and web plate after milling are $100.0 \times 150.0 \times 1.8 \text{ mm}^3$ and $3.0 \times 150.0 \times 90.0 \text{ mm}^3$, respectively. The laser power and welding speed are set as 1.5 kW and

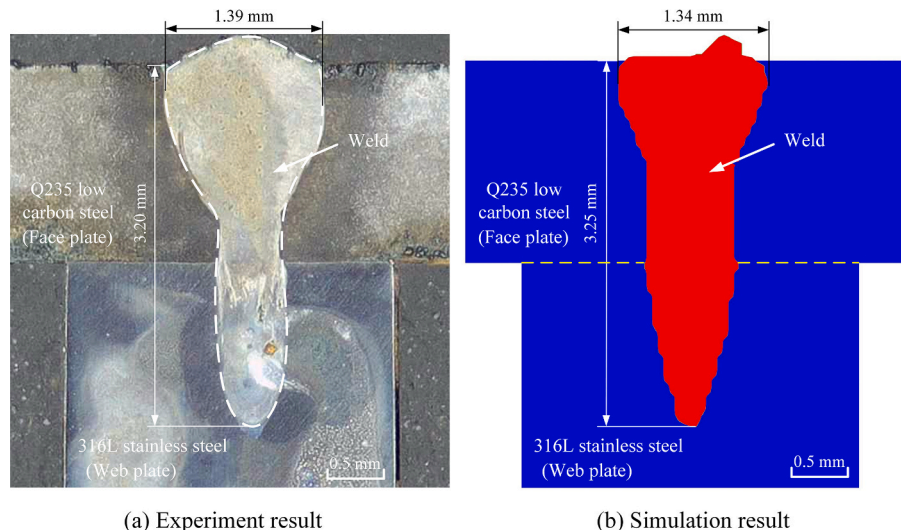


Fig. 8. Comparison of weld morphologies obtained from experiment and simulation of CLW.

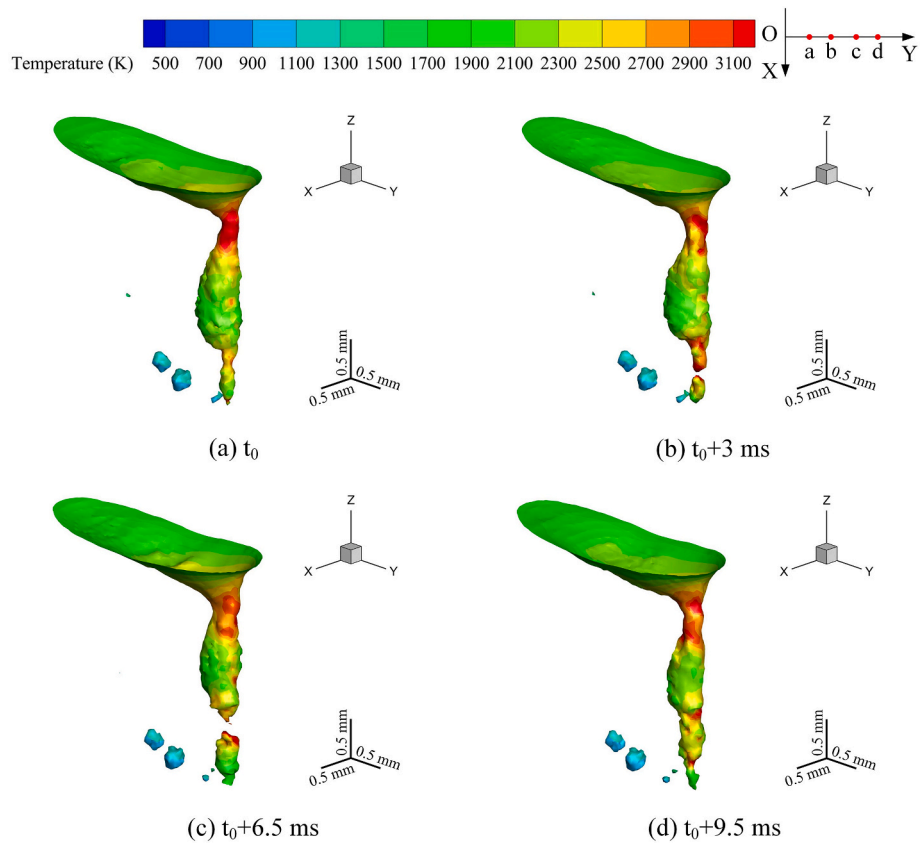


Fig. 9. Keyhole morphology evolution process at the steady stage of CLW.

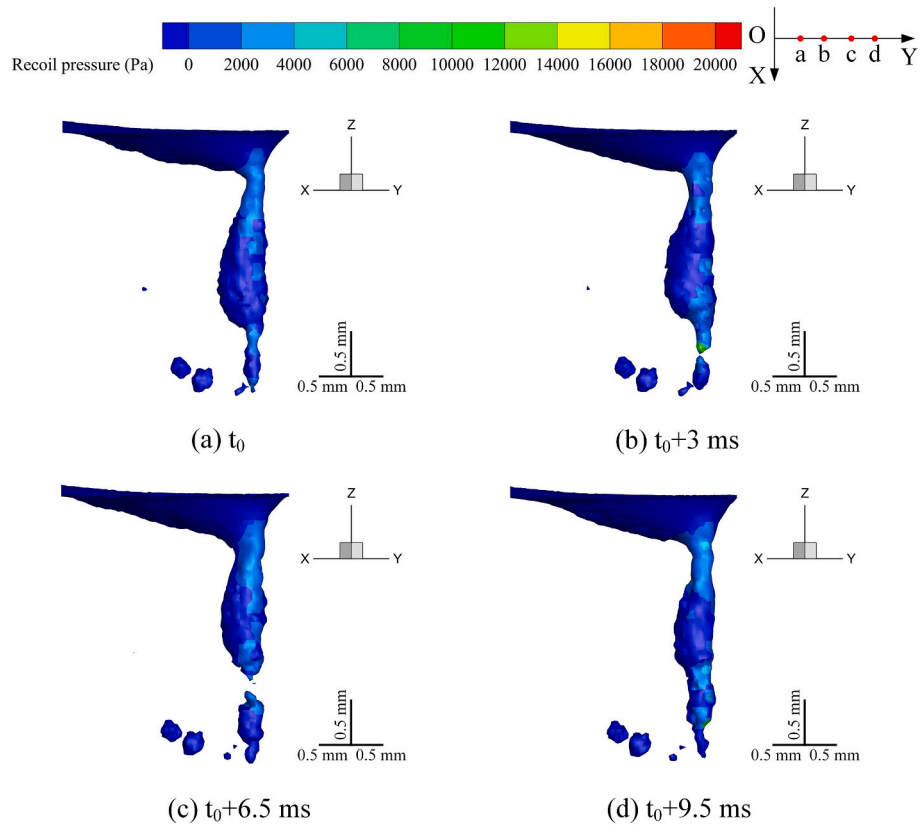


Fig. 10. Recoil pressure distributed on the keyhole wall at the steady stage of CLW.

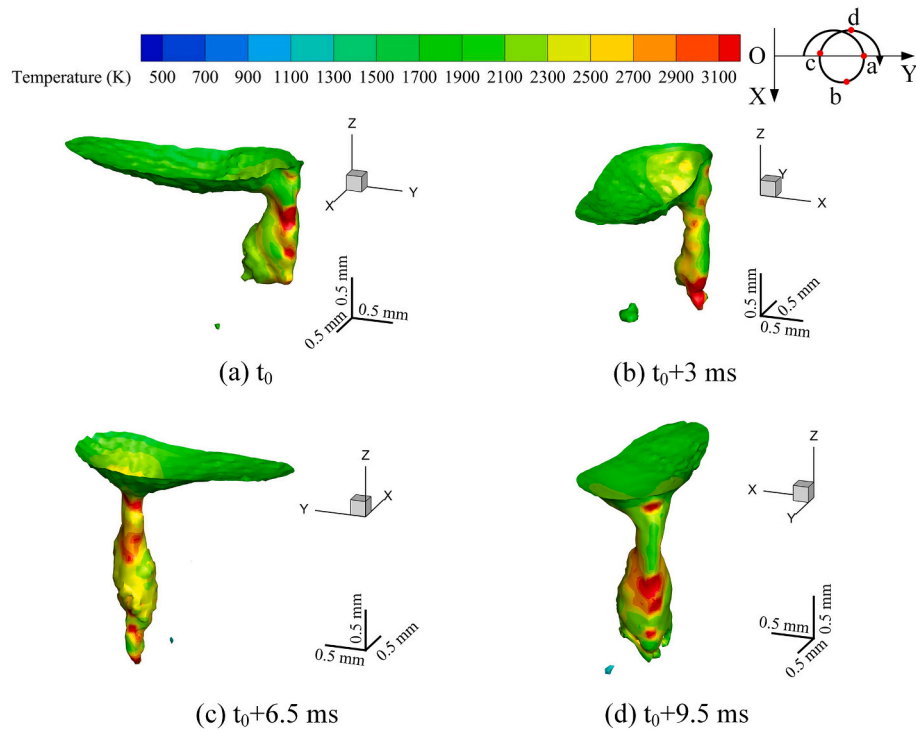


Fig. 11. Keyhole morphology evolution process at the steady stage of OLW.

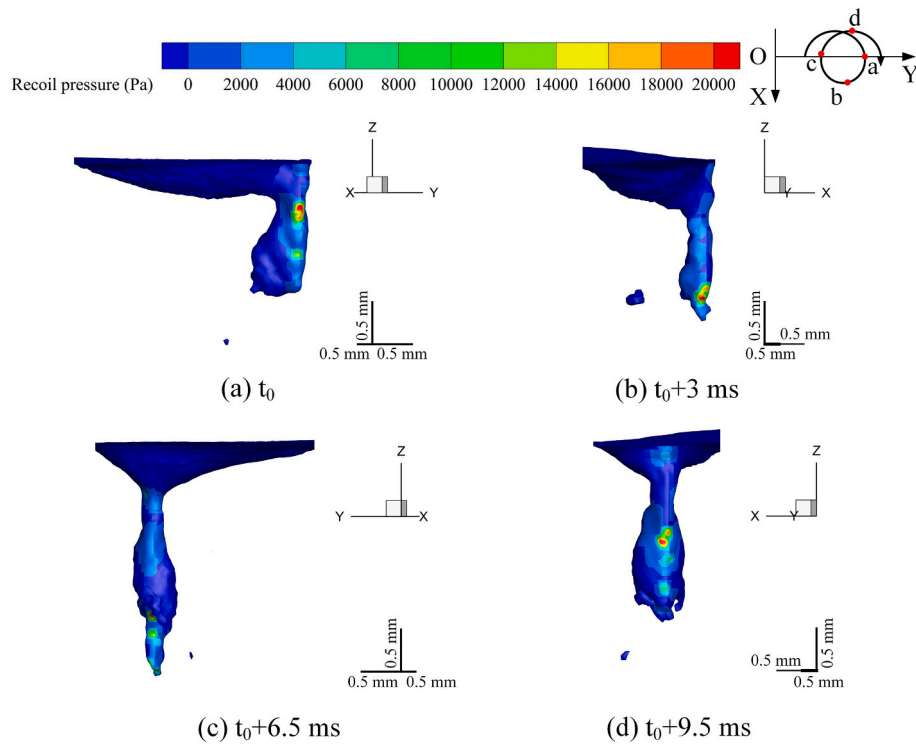


Fig. 12. Recoil pressure distributed on the keyhole wall at the steady stage of OLW.

1.2 m/min. The OLW is carried out by galvanometer scanner under the same process conditions as CLW, as shown in Fig. 2. The oscillation path is circular shape in clockwise in OLW. The oscillation amplitude and frequency are 0.8 mm and 80 Hz, respectively.

3. Numerical modeling

Some assumptions and simplifications are set to improve computational efficiency under the condition of ensuring the model accuracy. The effects of shielding gas and fixture on the laser welding process are not considered in the model. The thermal buoyancy is calculated by

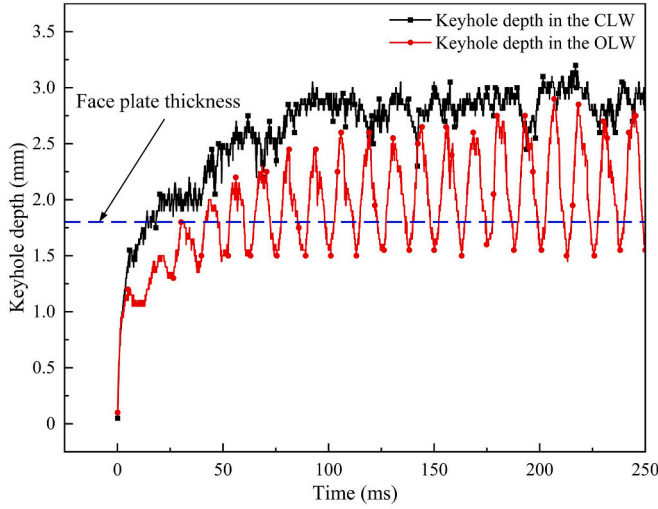


Fig. 13. The time-dependent curves of keyhole depths in CLW and OLW.

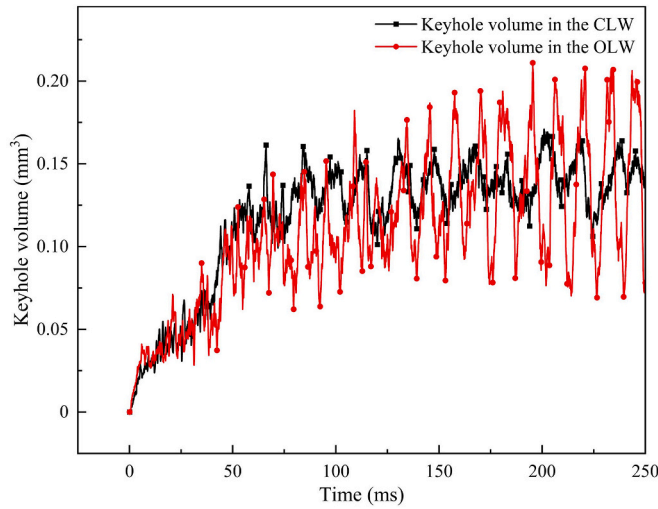


Fig. 14. The time-dependent curves of keyhole volumes in CLW and OLW.

Boussinesq approximation [24]. Governing equations are as follows:

Mass equation [25]

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{V}) = 0 \quad (1)$$

where ρ , t and $\vec{V}$ are density, time and velocity, respectively.

Momentum equation [25–28]

$$\frac{\partial (\rho \vec{V})}{\partial t} + \nabla \cdot (\rho \vec{V} \vec{V}) = -\nabla P + \nabla \cdot (\mu \nabla \vec{V}) - \frac{\mu}{K} \vec{V} + \rho \vec{g} - \rho \vec{g} \beta (T - T_l) + S_M \quad (2)$$

where P , μ , K , $\vec{g}$, β , T , T_l and S_M are pressure, viscosity, isotropic permeability, gravitational acceleration, thermal expansion coefficient, temperature, liquidus temperature and momentum source term, respectively.

Energy equation [25]

$$\frac{\partial (\rho H)}{\partial t} + \nabla \cdot (\rho \vec{V} H) = \nabla \cdot (k \nabla T) + S_E \quad (3)$$

where H , k and S_E are enthalpy, thermal conductivity and energy source term, respectively.

Volume of fluid (VOF) equation [28]

$$\frac{\partial F}{\partial t} + \vec{V} \cdot \nabla F = 0 \quad (4)$$

where F is VOF function.

The computational domain surfaces which belong to substrate surfaces are considered as wall boundary. The heat transfer phenomena on melt pool and keyhole free surfaces including laser energy absorption, thermal convection, thermal radiation and evaporation energy loss are described as [26]:

$$k \frac{\partial T}{\partial \vec{n}} = q_{absorbed} - h_{con}(T - T_0) - \epsilon \sigma (T^4 - T_0^4) - q_{evap} \quad (5)$$

where $\vec{n}$, $q_{absorbed}$, h_{con} , T_0 , ϵ , σ and q_{evap} are normal vector, absorbed energy, convection coefficient, ambient temperature, surface radiation emissivity, Stefan-Boltzmann constant and evaporation energy loss, respectively. The evaporation energy loss q_{evap} can be expressed as:

$$q_{evap} = W_{evap} H_v \quad (6)$$

where W_{evap} and H_v are evaporation rate and evaporation latent heat, respectively.

The energy loss by thermal convection and thermal radiation on wall boundary can be expressed as [26]:

$$k \frac{\partial T}{\partial \vec{n}} = -h_{con}(T - T_0) - \epsilon \sigma (T^4 - T_0^4) \quad (7)$$

The welding path of OLW is obtained by combining the welding path of CLW with laser beam oscillation path, as shown in Fig. 3. The absorbed energy $q_{absorbed}(x, y, z)$ by substrate can be characterized by the combination of double ellipsoid heat source and Gaussian surface heat source which can realize the rotation about the heat source central axis (Z_r) and the revolution along the laser beam oscillation path in OLW, as shown in Figs. 4 and 5. The double ellipsoid heat source is used to describe the irradiated energy $q_{metallic}(x, y, z)$ of metallic vapor and plasma on melt pool upper surface. The Gaussian surface heat source is used to express the irradiated energy $q_{laser}(x, y, z)$ of laser beam. During the calculation of ray multiple reflections process, the reflection ray is obtained based on incident ray and local normal vector of keyhole wall, as shown in Fig. 6. The energy of Fresnel absorption by keyhole wall is main part of absorbed energy $q_{absorbed}(x, y, z)$ by substrate. It is calculated by subtracting the lost laser energy $q_{lost}(x, y, z)$ due to the inverse bremsstrahlung absorption by metallic vapor and plasma [29,30]. The process of energy absorption in the laser welding can be calculated as [31–36]:

$$q_{absorbed}(x, y, z) = \eta q_{metallic}(x, y, z) + q_{laser}(x, y, z) - q_{lost}(x, y, z) \quad (8)$$

$$q_{metallic}(x, y, z) = \begin{cases} q_f(x, y, z) = \frac{6\sqrt{3}f_1f_rQ}{a_fbc\pi\sqrt{\pi}} \exp\left[-3\left(\frac{y_r^2}{a_f^2} + \frac{x_r^2}{b^2} + \frac{z_r^2}{c^2}\right)\right], & y_r > 0 \\ q_r(x, y, z) = \frac{6\sqrt{3}f_1f_rQ}{a_rbc\pi\sqrt{\pi}} \exp\left[-3\left(\frac{y_r^2}{a_r^2} + \frac{x_r^2}{b^2} + \frac{z_r^2}{c^2}\right)\right], & y_r \leq 0 \end{cases} \quad (9)$$

$$q_{laser}(x, y, z) = \frac{3f_2Q}{\pi r_0^2} \exp\left(-3\frac{x_r^2 + y_r^2}{r_0^2}\right) \quad (10)$$

$$f_1 + f_2 = 1 \quad (11)$$

$$f_f + f_r = 2 \quad (12)$$

$$\begin{cases} x_r = \Delta x \cos\theta - \Delta y \sin\theta \\ y_r = \Delta x \sin\theta + \Delta y \cos\theta \\ z_r = \Delta z \end{cases} \quad (13)$$

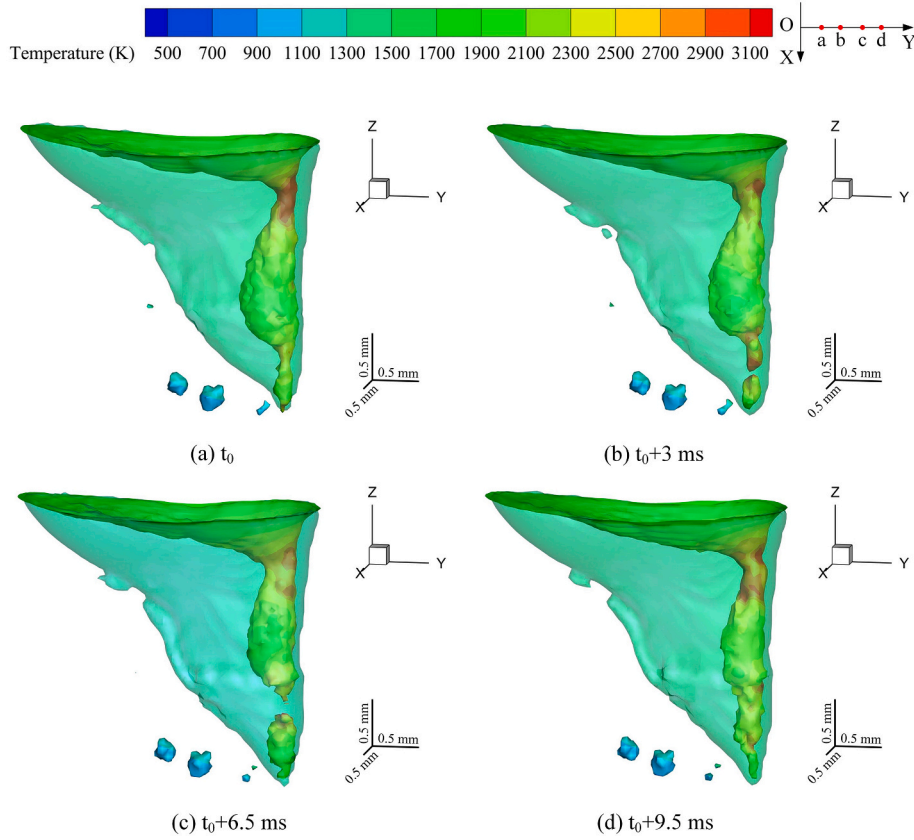


Fig. 15. Melt pool morphology evolution process at the steady stage of CLW.

$$\begin{cases} \Delta x = x - x_m \\ \Delta y = y - y_m \\ \Delta z = z - z_m \end{cases} \quad (14)$$

$$\begin{cases} x_m = x_0 + 0.5A\sin(2\pi ft) \\ y_m = y_0 + vt + 0.5A\cos(2\pi ft) \\ z_m = z_0 \end{cases} \quad (15)$$

$$\theta = 2\pi(ft - [ft]) + \frac{\pi}{2}\text{sgn}(Af) \quad (16)$$

where Q and r_0 are laser power and spot radius. η is absorptivity of energy of metallic vapor and plasma by substrate. f_1 and f_2 are energy distribution coefficients of double ellipsoid heat source and Gaussian surface heat source. f_f and f_r are energy distribution coefficients of front and back parts of double ellipsoid heat source. a_f , a_r , b and c are morphology parameters of double ellipsoid heat source, as shown in Fig. 4. x_0 , y_0 and z_0 are X, Y and Z coordinates of initial position of laser spot, respectively. v is welding speed. A and f are oscillation amplitude and frequency. $[ft]$ represents the largest integer not greater than ft in the floor function. $\text{sgn}(Af)$ is the signum function. $\text{sgn}(Af) = 0$ in CLW and $\text{sgn}(Af) = 1$ in OLW.

The keyhole is formed in melt pool under the condition of recoil pressure. In the numerical model, an improved formula with weighting according to volume fractions of dissimilar materials is proposed to calculate recoil pressure more accurately for dissimilar materials laser keyhole welding simulation. The recoil pressure P_r is calculated as [37]:

$$P_r = \frac{A_0 B_0}{\sqrt{T}} \exp \left[-\frac{(M_{a1} V_1 + M_{a2} V_2)(L_{v1} V_1 + L_{v2} V_2)}{N_a k_b T} \right] \quad (17)$$

where A_0 and B_0 are numerical coefficient and evaporation constant. M_{a1} and M_{a2} are average atomic masses of dissimilar materials. V_1 and V_2 are volume fractions of dissimilar materials in the cell which is

located on keyhole wall. L_{v1} and L_{v2} are latent heats of evaporation of dissimilar materials. N_a and k_b are Avogadro's number and Boltzmann's constant.

Surface tension P_σ on the free surfaces of melt pool and keyhole is described as [38]:

$$P_\sigma = \gamma \kappa \quad (18)$$

$$\gamma = \gamma_0 - A_T(T - T_l) \quad (19)$$

where γ , κ , γ_0 and A_T are surface tension coefficient, free surface curvature, surface tension at melting point and temperature coefficient of surface tension, respectively.

Marangoni stress τ_m is expressed as [39]:

$$\tau_m = \frac{\partial \gamma}{\partial T} \frac{\partial T}{\partial \vec{s}} \quad (20)$$

where $\vec{s}$ is tangential vector.

The mesh model with dimension of $4.0 \times 8.0 \times 4.8 \text{ mm}^3$ is adopted for numerical calculation, as shown in Fig. 7. The dimensions of face plate and web plate are $4.0 \times 8.0 \times 1.8 \text{ mm}^3$ and $3.0 \times 8.0 \times 2.0 \text{ mm}^3$, respectively. The dimensions of gas layers above face plate and on both sides of the web plate are $4.0 \times 8.0 \times 1.0 \text{ mm}^3$ and $0.5 \times 8.0 \times 2.0 \text{ mm}^3$, respectively. The thermophysical parameters are calculated based on material compositions by JMatPro and referenced from the previous literatures [40,41]. The laser welding process conditions for numerical calculation are kept the same as that in the welding experiments.

4. Results and discussion

4.1. Model validation

The CLW and OLW processes are calculated by the established

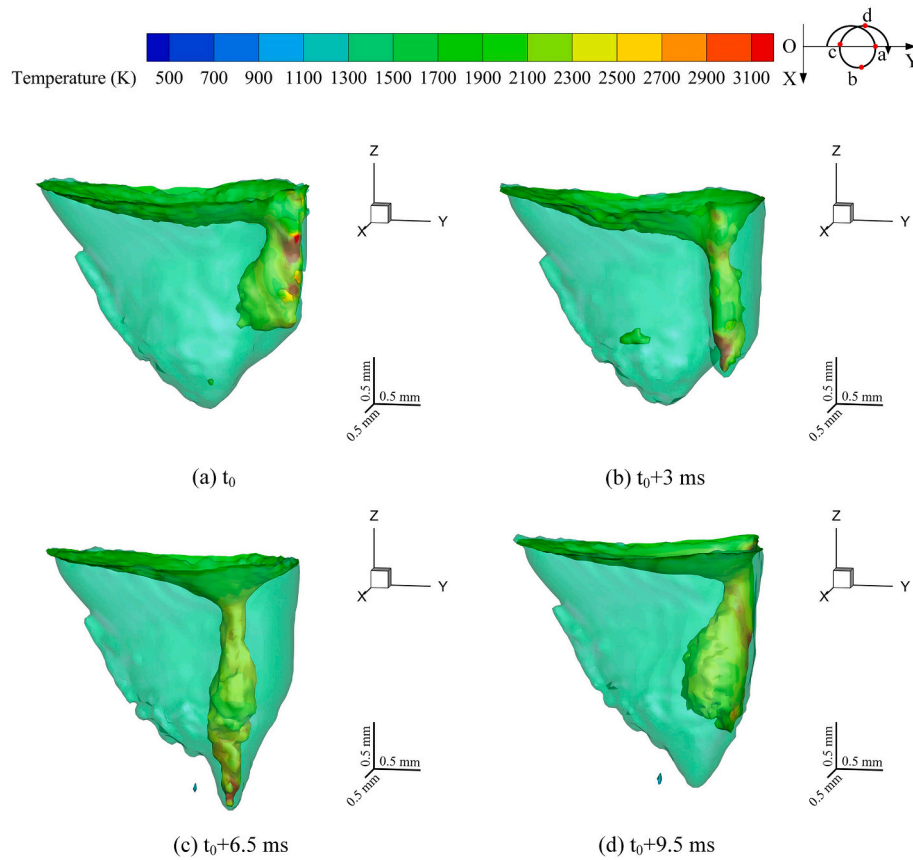


Fig. 16. Melt pool morphology evolution process at the steady stage of OLW.

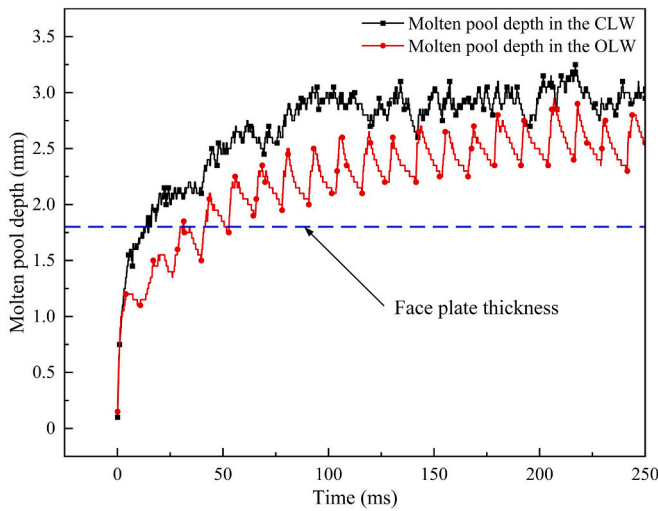


Fig. 17. The time-dependent curves of melt pool depths in CLW and OLW.

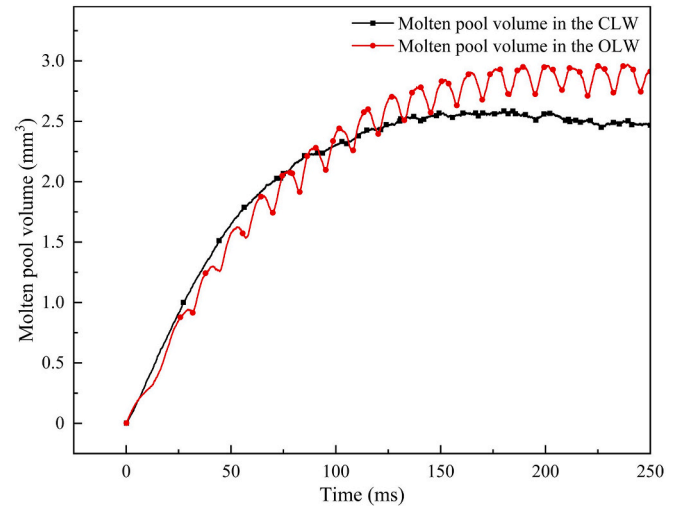


Fig. 18. Time-dependent curves of melt pool volumes in CLW and OLW.

numerical model. The comparison of weld morphologies obtained from experiment and simulation of CLW is shown in Fig. 8. It can be seen that the relative errors of weld depth and width are 1.56% and 3.60%, which indicates the validity of the numerical model.

4.2. The evolution process of keyhole morphology

The calculated keyhole morphology evolution process at the steady stage of CLW based on the numerical model is shown in Fig. 9. From Fig. 9, the keyhole morphology which is large in the center and narrow

in the upper and bottom parts is kept almost unchanged after CLW reaching the steady stage. The morphology characteristics of keyhole can cause keyhole collapse easily and then bubbles are generated in melt pool, as shown in Figs. 9(b) and (c). The high temperature regions are distributed on the wall of the upper and bottom parts of keyhole, which means that laser beam mainly radiates on the wall of the upper part of keyhole to keep keyhole opening and some rays are able to pass through keyhole and reach the keyhole bottom part. The recoil pressure distributed on the keyhole wall in CLW is shown in Fig. 10. From Fig. 10, the regions with high recoil pressure correspond to the regions with high

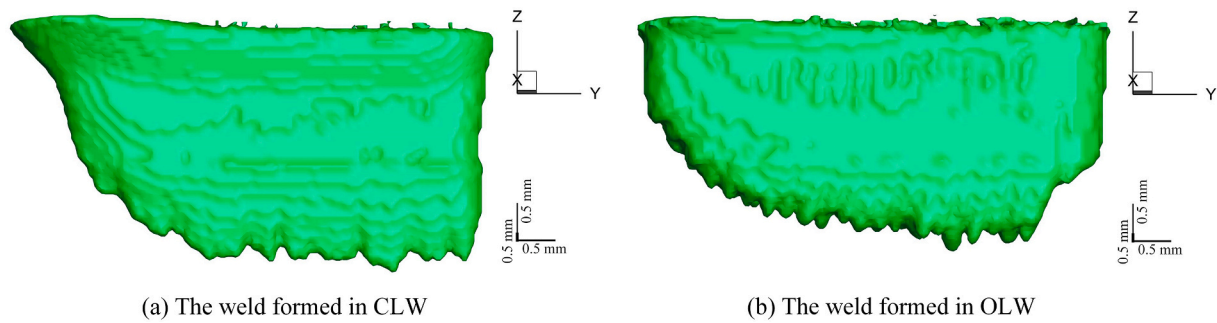


Fig. 19. Comparison of weld morphologies formed in CLW and OLW.

temperature shown in Fig. 9. The positions of high recoil pressure regions on the keyhole wall are kept almost unchanged at the steady stage of CLW. There are some bubbles formed near the keyhole bottom part, which is due to keyhole collapse.

The keyhole morphology evolution process at the steady stage of OLW is shown in Fig. 11. From Fig. 11, the keyhole morphology variation is more significant at the steady stage of OLW than that of CLW. When laser beam moves from the front of melt pool to the rear, the keyhole depth is increased and the high temperature regions are transferred from the front wall relative to actual movement direction of keyhole to the bottom wall of keyhole, as shown in Figs. 11(a) and (b). The keyhole depth is decreased and high temperature regions are transferred from the bottom wall of keyhole to the front wall relative to actual movement direction of keyhole when laser beam moves from the rear of melt pool to the front again, as shown in Figs. 11(c) and (d). The lower part of keyhole shown in Figs. 11(a) and (d) is larger than that shown in Figs. 11(b) and (c). This is due to the fact that as laser beam moves from the rear of melt pool to the front, the angle between laser welding direction and laser beam oscillation direction is an acute angle, which indicates that actual linear speed of oscillating laser beam is increased in OLW. The left space caused by the moving keyhole behind oscillating laser beam is hard to be filled by molten metal in melt pool in time during high speed movement process of oscillating laser beam. The recoil pressure distributed on the keyhole wall in OLW is shown in Fig. 12. From Fig. 12, the regions with high recoil pressure are generally consistent with the regions of high temperature shown in Fig. 11. The moving trend of high recoil pressure regions is almost kept the same as that of high temperature regions.

The time-dependent curves of keyhole depths in CLW and OLW are shown in Fig. 13. From Fig. 13, the keyhole depth is increased rapidly at first, then increased slowly and finally kept in a relatively steady state with fluctuations in the small range in CLW, while that in OLW is fluctuant over time in a large fluctuation range and it is increased firstly and then kept in a steady state in continuous fluctuations. In addition, the fluctuation of keyhole depth in OLW shows periodic characteristics and fluctuation period is almost the same as laser beam oscillation period. During an oscillation period of OLW, the keyhole depth is small when laser beam radiates on the upper surface of the front of melt pool, while that is becoming larger when laser beam moves towards the rear of melt pool. This is because the deposition of laser energy during the previous welding process makes the temperature in the rear of melt pool to be higher than that in the front of melt pool. The evaporation process of molten metal is more drastic and the generated recoil pressure is stronger, which causes that laser beam radiates directly on the bottom wall of keyhole to increase keyhole depth as laser beam moves towards the rear of melt pool. The keyhole average depth in OLW is smaller than that in CLW. The keyhole always penetrates through the face plate after about 18 ms in CLW, while the keyhole penetrates into and out of the web plate once in each oscillation period after about 30 ms in OLW. The time-dependent curves of keyhole volumes in CLW and OLW are shown in Fig. 14. The upper surface of melt pool is usually lower than the upper

surface of face plate in laser welding possibly due to substrate material loss by evaporation and surface tension effect. The volume of grids which are dominated by metallic vapor and plasma in the region surrounded by melt pool in the computational domain is regarded as the keyhole volume. From Fig. 14, the keyhole volumes in CLW and OLW are both increased firstly and then kept in the steady state in continuous fluctuations. The fluctuation period of keyhole volume in the OLW is nearly the same as laser beam oscillation period and is smaller than that in CLW. The fluctuation range of keyhole volume in OLW is larger than that in CLW. When laser beam moves from the front of melt pool to the rear in each oscillation period of OLW, the keyhole volume is kept increasing. This is because recoil pressure is stronger when laser beam is located at the rear of melt pool.

4.3. The evolution process of melt pool morphology

The melt pool morphologies evolution processes in CLW and OLW are shown in Figs. 15 and 16. From Fig. 15, the melt pool morphology with large length and depth and the keyhole relative position in melt pool are both kept almost unchanged after CLW reaching the steady stage. From Fig. 16, the melt pool morphology and keyhole relative position in melt pool show obvious variations during an oscillation period after OLW reaching the steady stage. The keyhole relative position in melt pool is kept changing with laser beam oscillation. The melt pool length is decreased and the melt pool depth is increased when laser beam moves from the front of melt pool to the rear, as shown in Figs. 16 (a), (b) and (c). The melt pool length is increased and the melt pool depth is decreased when laser beam moves from the rear of melt pool to the front, as shown in Figs. 16(c) and (d). The time-dependent curves of melt pool depths in CLW and OLW are shown in Fig. 17. Whether in CLW or OLW, the variation trend of melt pool depth is kept almost consistent with that of keyhole depth and the fluctuation range of melt pool depth is smaller than that of keyhole depth. The time when the melt pool depth always exceeds the face plate thickness in CLW is about 16 ms, while that in OLW is about 52 ms. The time-dependent curves of melt pool volumes in CLW and OLW are shown in Fig. 18. From Fig. 18, the melt pool volumes in CLW and OLW are both increased slowly at first and then kept in the relatively steady state. The melt pool volume in OLW shows periodic fluctuation characteristics and fluctuation period is nearly the same as laser beam oscillation period. The melt pool volume is increased with laser beam moving from the rear of melt pool to the front during an oscillation period of OLW. It is because the laser energy is mainly used to vaporize molten metal more drastically and generate larger recoil pressure when laser beam radiates on the upper surface of the rear of melt pool, while that is mainly used to heat and melt more solid metal when laser beam radiates on the upper surface of the front of melt pool. The fluctuation range of melt pool volume in CLW is smaller than that in OLW. Whether in CLW or OLW, the fluctuation range of melt pool volume is smaller than that of keyhole volume. The melt pool volume in OLW is always larger than that in CLW after about 133 ms. The comparison of weld morphologies formed in CLW and OLW is

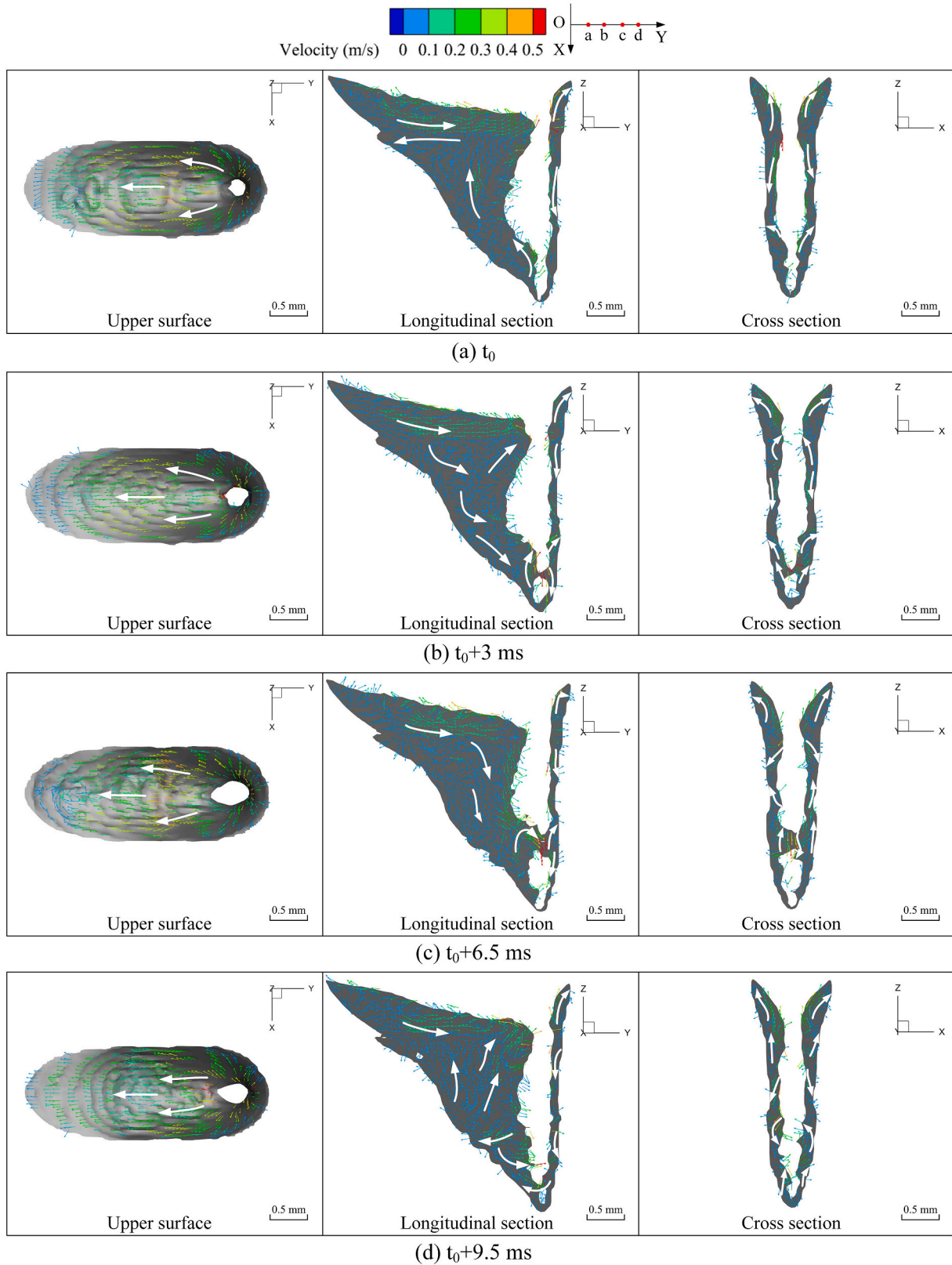


Fig. 20. Flow fields on the melt pool upper surface and in the melt pool longitudinal section and cross section in CLW.

shown in Fig. 19. From Fig. 19, the weld formed in CLW shows several irregular bumps and depressions in the bottom part, while that formed in OLW periodically demonstrates the relatively regular bumps in the bottom part.

4.4. The flow behaviors of molten metal

The flow fields on the melt pool upper surface and in the melt pool longitudinal section and cross section in CLW and OLW are shown in Figs. 20 and 21, respectively. From the upper surface of melt pool in

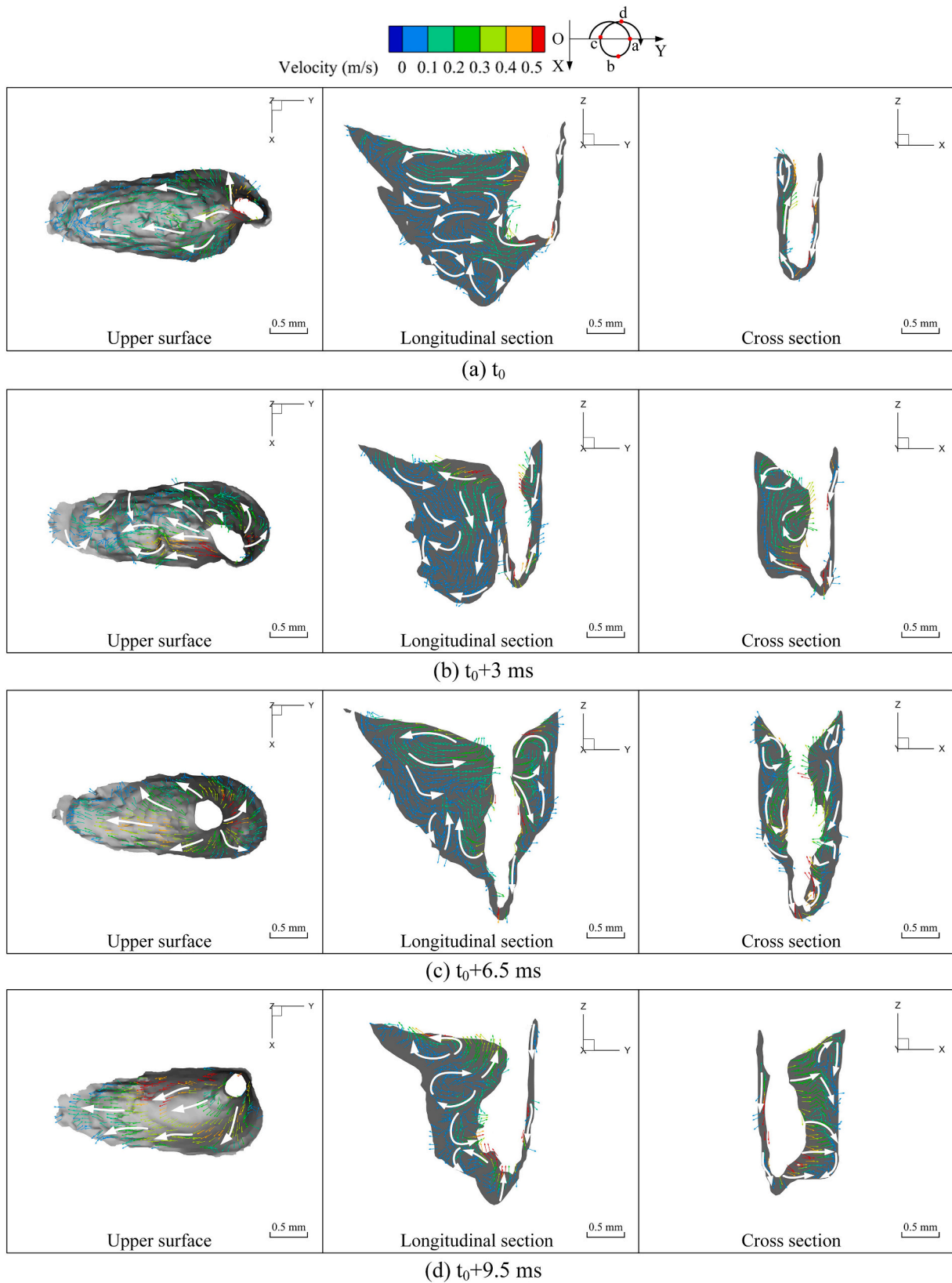


Fig. 21. Flow fields on the melt pool upper surface and in the melt pool longitudinal section and cross section in OLW.

Figs. 20 and 21, the molten metal on the melt pool upper surface flows from the keyhole center to the rear of melt pool in CLW and OLW. The flow direction of molten metal is more complicated in OLW than that in CLW at the same time. From the longitudinal section of melt pool in

Fig. 20, the difference in flow velocity magnitude of molten metal in different regions of melt pool longitudinal section at the steady stage of CLW is obvious. The regions with high flow velocity magnitude are mainly distributed on the melt pool upper part and surrounding regions

of keyhole wall. The area of regions with high flow velocity magnitude is small and flow velocity magnitude of molten metal in most regions is low. From the longitudinal section of melt pool in Fig. 21, comparing with the steady stage of CLW, the distribution of molten metal flow velocity magnitude is more uniform and area of regions with high flow velocity magnitude is increased in the melt pool longitudinal section during an oscillation period of OLW. Comparing with CLW, the flow behaviors of molten metal is more complicated in OLW. Many vortices with large area are generated in melt pool under the oscillating laser beam stirring effect, which can promote bubbles escape and thus reduce pore defect. From the cross section of melt pool in Figs. 20 and 21, the melt pool morphology, flow direction and flow velocity magnitude of molten metal in most regions of melt pool cross section are all approximately symmetric about the keyhole at the steady stage of CLW, while that all change significantly in oscillating laser beam movement process in OLW. Comparing with the steady stage of CLW, the number of vortices, the flow velocity magnitude of molten metal and area of regions with high flow velocity magnitude are all increased obviously in the melt pool cross section under the stirring effect of oscillating laser beam in OLW, which indicates that the bubbles are more likely to escape out of melt pool by laser beam oscillation.

5. Conclusion

A novel numerical model based on a combined heat source and an improved recoil pressure formula is developed to explore keyhole and melt pool evolution processes in CLW and OLW of dissimilar materials stake welded T-joints. The keyhole and melt pool morphologies evolution processes and molten metal flow behaviors in CLW and OLW are analyzed. The conclusions are summarized as follows:

- (1) The keyhole is large in the center and narrow in the upper and bottom parts during the steady stage of CLW. The keyhole depths and volumes in CLW and OLW are all increased firstly and then kept in a steady state. Both of keyhole depth and volume in OLW show periodic fluctuation characteristics with approximately the same period as laser beam oscillation. Comparing with CLW, the keyhole average depth is smaller, fluctuation period of keyhole volume is smaller, and fluctuation ranges of keyhole depth and volume are both larger in OLW.
- (2) Both of melt pool length and depth are large at the steady stage of CLW. Whether in CLW or OLW, the variation trend of melt pool depth almost corresponds to that of keyhole depth and fluctuation range of melt pool depth is smaller than that of keyhole depth. The melt pool volume in OLW shows periodic fluctuation characteristics with roughly the same period as the laser beam oscillation. The morphology characteristics in the bottom of weld formed in OLW is more regular than that formed in CLW.
- (3) Comparing with CLW, the distribution of molten metal flow velocity magnitude is more uniform and area of regions with high flow velocity magnitude is increased at the steady stage of OLW. The molten metal flow direction is more complicated and many vortices with large area are formed in melt pool under oscillating laser beam stirring effect, which can help bubbles escape from melt pool.

CRediT authorship contribution statement

Yuewei Ai: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Chenglong Ye:** Writing – review & editing, Writing – original draft, Visualization, Validation, Data curation, Formal analysis, Methodology. **Jiabao Liu:** Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation. **Ming Zhou:** Writing – review & editing, Validation, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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